

Pattern Formation in a Model of Acute Inflammation

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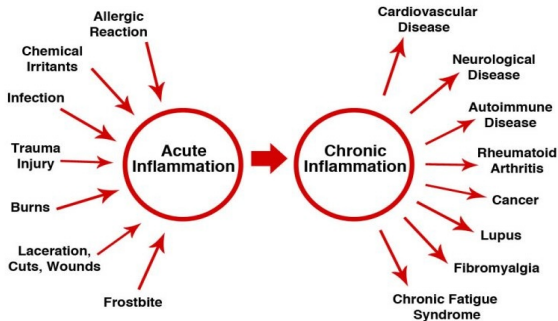
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A dysregulation of the inflammatory response can give rise to a vast range of chronic and debilitating diseases



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4. Show some possible 2D patterns

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The two types of signalling molecules (chemokines and cytokines) are both produced by macrophages; the first is an attractor for immune cells (it activates and recruits other macrophages), the second is an inhibitor (quenches the inflammatory activity).

Biological Background

Chemotaxis: is a biological process in which organisms or cells move towards or away from chemical stimuli. It is a universal phenomenon observed in bacteria, single-celled eukaryotes, and multicellular organisms.

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How it works:

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Response: binding of attractants or repellents triggers intracellular signals

Action: this alters cellular motility, affecting cytoskeletal dynamics and adhesion

The Keller Segel model

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$$m_t = \nabla \cdot (k_1(m, c) \nabla m - k_2(m, c) \nabla c),$$

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Handle with care: existence and regularity of solutions depend on the chemotactic term $k_2(m, c)$: for many natural choices the solutions could blow up in two dimensions.

ex: for $k_2(m, c) = m$ the solutions blow up in 2D domains.

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It has been shown that the inclusion of a **limited-growth chemotactic term** or a **logistic-type reaction term** has a blow-up-inhibiting effect.

Our modified version of the model

We propose the following three-variable PDE system:

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Activated macrophages

$m(x, t)$

$$m_t = \nabla \cdot \left(D_m \nabla m - \frac{\chi m}{(1+\alpha c)^2} \nabla c \right)$$

Pro-inflammatory chemokines

$c(x, t)$

$$c_t = D_c \Delta c - c + \frac{m}{1+\beta a^\rho}$$

Anti-inflammatory cytokines

$a(x, t)$

$$a_t = \left(D_c \Delta a - a + \frac{m}{1+\beta a^\rho} \right) \frac{1}{\tau}$$

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D_m and D_c are the diffusion coefficients, respectively for macrophages and for the signalling molecules that they produce.

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χ represents the maximal chemotactic rate for the macrophages.

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The constants β and ρ modulate the inhibitory effect of anti-inflammatory cytokines.

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Macrophages are attracted by the chemokines signal, but the saturation of cell receptors leads to a decrease in the rate of attraction. Following the principle of chemotaxis, the macrophages detect the chemokine signal and migrate towards higher concentrations of chemokine, moving up the chemical gradient.

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The term $-c$ (respectively $-a$) indicates the natural decay or removal rate of chemokines (respectively cytokines) from the environment. The fraction $\frac{m}{1+\beta a^\rho}$ is related to the production and decay of signalling molecules. Both the production and decay of the pro-inflammatory chemokine and the anti-inflammatory cytokine are represented by identical terms, but the anti-inflammatory cytokine's response is governed by a timescale $\tau > 1$, considered as a variable parameter.

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The system:

$$\begin{aligned}m_t &= \nabla \cdot \left(D_m \nabla m - \frac{\chi m}{(1+\alpha c)^2} \nabla c \right), \\c_t &= D_c \Delta c - c + \frac{m}{1+\beta a^\rho}, \\a_t &= \left(D_c \Delta a - a + \frac{m}{1+\beta a^\rho} \right) \frac{1}{\tau}.\end{aligned}$$

becomes precisely a two-variable *Keller-Segel* type model:

$$\begin{aligned}m_t &= \nabla \cdot (k_1(m, c) \nabla m - k_2(m, c) \nabla c), \\c_t &= D_c \nabla^2 c - k_3(c) c + m f(c).\end{aligned}$$

when the parameter τ has value 1, and in this case the concentrations of chemokine c and cytokine a exhibit identical dynamical behaviour. The inclusion of receptor saturation in the chemotactic term **ensures bounded solutions** and **prevents explosive growth**.

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In finite domains, for simulation purposes, the total quantity of macrophages will be denoted as:

$$M = \int_{\Omega} m(x, t) dx.$$

Instead, in the cases in which the domain is boundless, there exists an endless number of macrophages.

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When $\rho = 1$ the solution becomes

$$a_0 = \frac{-1 + \sqrt{1 + 4\beta m_0}}{2\beta}.$$

We reparametrize the homogeneous solution using a_0 in place of m_0 :

$$P_0 := (m_0, c_0, a_0) = (a_0(1 + \beta a_0^\rho), a_0, a_0).$$

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Important: The total macrophage population is conserved in our model, implying that once we fix the total population M , we also set m_0 .

Linearization

To study the stability of the homogeneous states, we **linearize** the system:

$$\frac{\partial}{\partial t} \begin{bmatrix} m \\ c \\ a \end{bmatrix} = \begin{bmatrix} D_m & \frac{-\chi a_0(1+\beta a_0^\rho)}{(1+\alpha a_0)^2} & 0 \\ 0 & D_c & 0 \\ 0 & 0 & \frac{D_c}{\tau} \end{bmatrix} \Delta \begin{bmatrix} m \\ c \\ a \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ \frac{1}{1+\beta a_0^\rho} & -1 & \frac{-\beta \rho a_0^\rho}{1+\beta a_0^\rho} \\ \frac{1}{\tau(1+\beta a_0^\rho)} & 0 & -\frac{1}{\tau} \left(\frac{(1+\beta(1+\rho)a_0^\rho)}{1+\beta a_0^\rho} \right) \end{bmatrix} \begin{bmatrix} m \\ c \\ a \end{bmatrix}.$$

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Looking for solutions of the form $(\hat{m}, \hat{c}, \hat{a})(x, t) = e^{\lambda t} \Phi_k(x)$, we substitute and obtain the eigenvalue problem:

$$\lambda \begin{bmatrix} \hat{m} \\ \hat{c} \\ \hat{a} \end{bmatrix} = A(k) \begin{bmatrix} \hat{m} \\ \hat{c} \\ \hat{a} \end{bmatrix},$$

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with

$$A(k) = \begin{bmatrix} -k^2 D_m & \frac{k^2 \chi a_0(1+\beta a_0^\rho)}{(1+\alpha a_0)^2} & 0 \\ \frac{1}{1+\beta a_0^\rho} & -1 - k^2 D_c & \frac{-\beta \rho a_0^\rho}{1+\beta a_0^\rho} \\ \frac{1}{\tau(1+\beta a_0^\rho)} & 0 & -\frac{1}{\tau} \left(\frac{(1+\beta(1+\rho)a_0^\rho)}{1+\beta a_0^\rho} + k^2 D_c \right) \end{bmatrix}.$$

Theorem (V. Giunta, M.C. Lombardo, M. Sammartino)

If $k = 0$, then the homogeneous equilibrium P_0 is linearly stable

Without diffusion and chemotaxis, P_0 is stable.

Theorem (V. Giunta, M.C. Lombardo, M. Sammartino)

If $\chi = 0$, then the homogeneous equilibrium P_0 is linearly stable

The linear diffusion is not able to destabilize P_0 . Chemotaxis is the only potentially destabilizing mechanism.

Hereafter, **we shall assume $\chi \neq 0$**

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if $\operatorname{Re}(\lambda) < 0$ for all λ and all k , then the homogeneous state remains stable.

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We have to analyze the eigenvalues of $A(k)$

Study of eigenvalues

In order to analyze the eigenvalues of the linearized system's matrix $A(k)$, Gershgorin's theory can be used. Let's recall the following:

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Theorem (Gershgorin)

Let $A \in \mathbb{R}^{n \times n}$ (or $\mathbb{C}^{n \times n}$), then:

$$\text{Spec}(A) \subseteq \bigcup_{i=1}^n K_i \subseteq \mathbb{C}$$

where $K_i := \{\alpha \in \mathbb{C} \mid |a_{ii} - \alpha| \leq \sum_{j \neq i} |a_{ij}|\}$ is called the i -th Gershgorin disc.

For the matrix $A(k)$, the Gershgorin discs corresponding to rows $i = 1, 2, 3$ are defined as:

$$i = 1 : \quad |\lambda + k^2 D_m| \leq \frac{k^2 \chi a_0 (1 + \beta a \rho_0)}{(1 + \alpha a_0)^2},$$

$$i = 2 : \quad |\lambda + 1 + k^2 D_c| \leq 1,$$

$$i = 3 : \quad \left| \lambda + \frac{1}{\tau} [1 + \beta(1 + \rho)a\rho_0] + k^2 D_c \right| \leq \frac{1}{\tau(1 + \beta a \rho_0)}.$$

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Considering $\rho = 1$ in our computational experiments, discs for $i = 2, 3$ never contain 0, regardless of any parameter value. However, for $i = 1$, the inclusion of 0 occurs if and only if:

$$\frac{\chi a_0 (1 + \beta a \rho_0)}{D_m (1 + \alpha a_0)^2} > 1.$$

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$$i = 3 : \quad \left| \lambda + \frac{1}{\tau} [1 + \beta(1 + \rho)a\rho_0] + k^2 D_c \right| \leq \frac{1}{\tau(1 + \beta a \rho_0)}.$$

Considering $\rho = 1$ in our computational experiments, discs for $i = 2, 3$ never contain 0, regardless of any parameter value. However, for $i = 1$, the inclusion of 0 occurs if and only if:

$$\frac{\chi a_0 (1 + \beta a \rho_0)}{D_m (1 + \alpha a_0)^2} > 1.$$

Lower a_0 (and hence m_0) values or **reduced** $\frac{\chi}{D_m}$ (weaker chemoattraction) $\rightarrow$ **NO** pattern formation.

To proceed further in our examination, we are going to utilize Routh-Hurwitz theory.

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Routh–Hurwitz stability criterion

Given the polynomial

$$P(\lambda) = \lambda^n + a_1\lambda^{n-1} + \cdots + a_{n-1}\lambda + a_n,$$

where the coefficients a_i are real constants, $i = 1, \dots, n$, define the n Hurwitz matrices using the coefficients a_i of the characteristic polynomial:

$$H_1 = [a_1], \quad H_2 = \begin{bmatrix} a_1 & 1 \\ a_3 & a_2 \end{bmatrix}, \quad H_3 = \begin{bmatrix} a_1 & 1 & 0 \\ a_3 & a_2 & a_1 \\ a_5 & a_4 & a_3 \end{bmatrix},$$

Routh–Hurwitz stability criterion

and in general

$$H_n = \begin{bmatrix} a_1 & 1 & 0 & \cdots & 0 \\ a_3 & a_2 & a_1 & \cdots & 0 \\ a_5 & a_4 & a_3 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & a_n \end{bmatrix},$$

where $a_j = 0$ if $j > n$. All the roots of the polynomial $P(\lambda)$ are negative or have negative real part if and only if the determinants of all Hurwitz matrices are positive:

$$\det H_j > 0, \quad j = 1, 2, \dots, n.$$

In particular, for $n = 2$, this is equivalent to $a_1 > 0$, $a_1 a_2 - a_3 > 0$ and $a_3 > 0$.

Let $\mathcal{P}(\lambda) = \lambda^3 + N(k)\lambda^2 + P(k)\lambda + Q(k)$ be the characteristic polynomial. From Routh-Hurwitz criterion, all roots have negative real part if and only if:

a) $N > 0$ (in our case, **always**),

b) $Q > 0$ and

c) $R := NP - Q > 0$.

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χ_T is the **Turing bifurcation threshold**

Similarly, we can negate c) (i.e. asking for $R < 0$). By doing so, we find that:

For τ sufficiently large, $\exists \chi_H > 0$ such that for $\chi > \chi_H$

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Note also that *there could exist* values of k for which $Q(k)R(k) < 0$.

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Note also that *there could exist* values of k for which $Q(k)R(k) < 0$. This asymmetry leads to a **bifurcation diagram**.

Bifurcation diagram

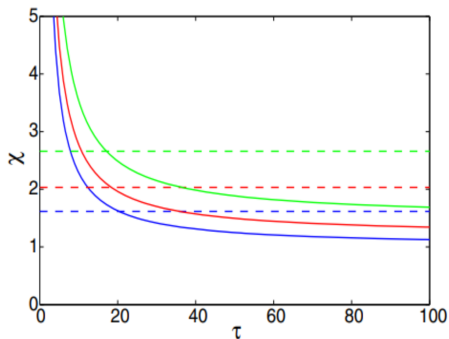


Figure 1: Turing and Hopf bifurcation curves (χ_T dashed, χ_H solid) for different values of m_0 . Parameters are as in (8) and $m_0 = 10$ (*blue* curves), 20 (*red*), or 40 (*green*).

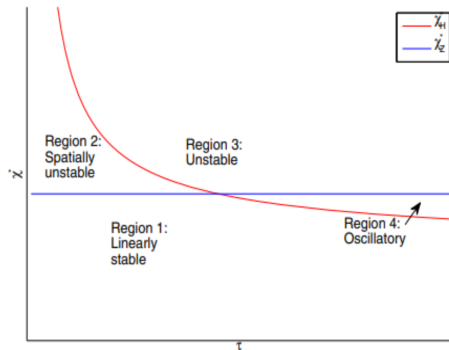


Figure 2: Scheme of the regions of stability determined by a pair of Turing and Hopf bifurcation curves.

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Discretization with a finite-difference scheme: $i = 0, \dots, 199$, $f_i = \frac{\chi m_i}{(1 + \alpha c_i)^2}$

$$\frac{d}{dt}m_i = \frac{D_m}{\Delta x^2}(m_{j+1} + m_{j-1} - 2m_j) - \frac{(f_{j+1} + f_j)(c_{j+1} - c_j) - (f_j + f_{j-1})(c_j - c_{j-1})}{2\Delta x^2},$$

$$\frac{d}{dt}c_i = \frac{D_c}{\Delta x^2}(c_{j+1} - 2c_j + c_{j-1}) - c_j + \frac{m_i}{1 + \beta a_j^p},$$

$$\frac{d}{dt}a_i = \left(\frac{D_c}{\Delta x^2}(a_{j+1} - 2a_j + a_{j-1}) - a_j + \frac{m_i}{1 + \beta a_j^p} \right) \frac{1}{\tau}$$

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Periodic conditions:

$$(m_{200+j}, c_{200+j}, a_{200+j})(t) = (m_j, c_j, a_j)(t) \text{ for } j = -1, 0.$$

Initial conditions:

$$m(x, 0) = m_0 = 10, \quad a(x, 0) = 5, \quad c(x, 0) = 5 + \xi(x)$$

where $\xi(x)$ indicates a *uniformly distributed random variable* in $(-0.1, 0.1)$.

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Selecting values of χ and τ in different areas of the bifurcation diagram, we are able to notice very different behaviours of solutions.

Representation of $m(x, t)$

$$\chi = 1.75, \tau = 10$$

This choice of (χ, τ) lies in the region of Turing instability and shows an expected stationary pattern of bumps.

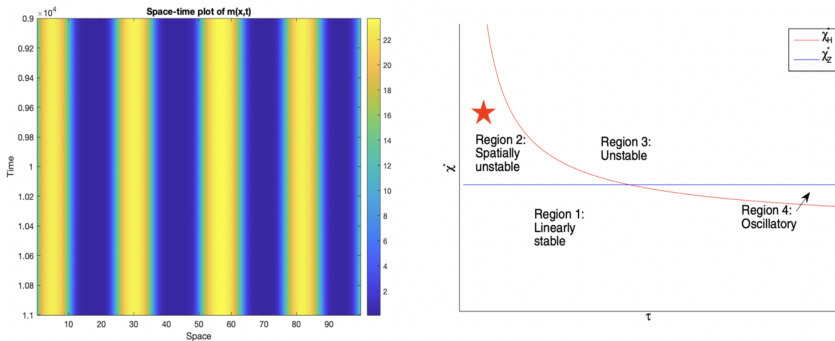


Figure 3: Representation of $m(x, t)$ for $(\chi, \tau) = (1.75, 10)$ and position in the bifurcation diagram.

$$\chi = 2.5, \tau = 30$$

This choice of (χ, τ) lies in the region of both Turing and Hopf instability, where both Q and R are negative for some values of K (not necessarily the same). The local stability analysis makes no predictions, but we again observe a stationary pattern.

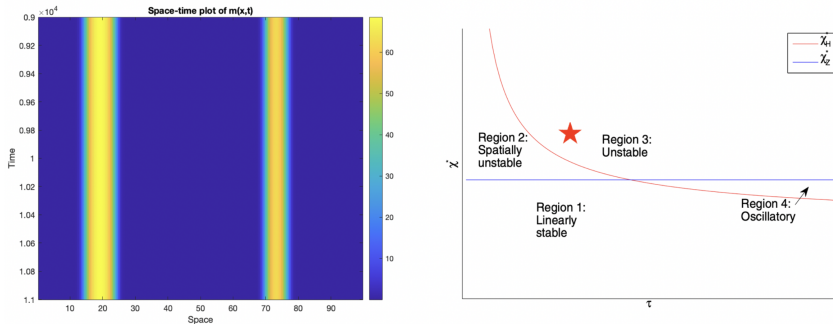


Figure 4: Representation of $m(x, t)$ for $(\chi, \tau) = (2.5, 30)$ and position in the bifurcation diagram.

$$\chi = 1.5, \tau = 100$$

This choice of (χ, τ) lies in the region of Hopf instability. As predicted by the linear analysis, small amplitude spatially periodic structure with time-periodic oscillations appears.

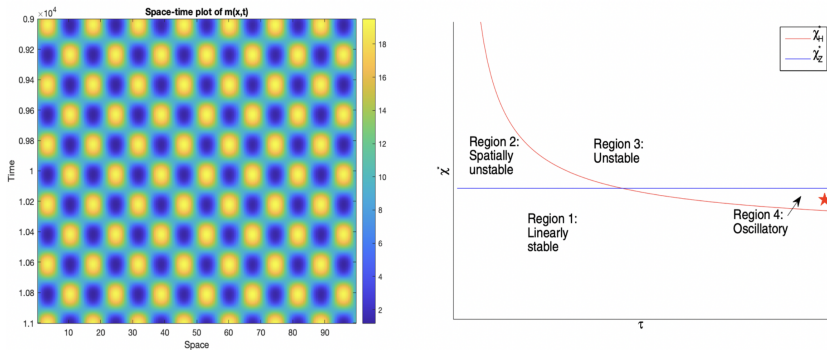


Figure 5: Representation of $m(x, t)$ for $(\chi, \tau) = (1.5, 100)$ and position in the bifurcation diagram.

$$\chi = 2, \tau = 100$$

This choice of (χ, τ) lies again in the region of both Turing and Hopf instability, but we're considering a much higher value of τ : in this case, the pattern has the form of travelling bumps.

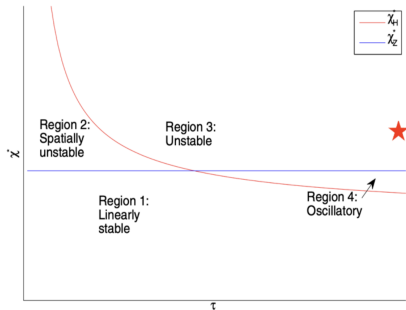
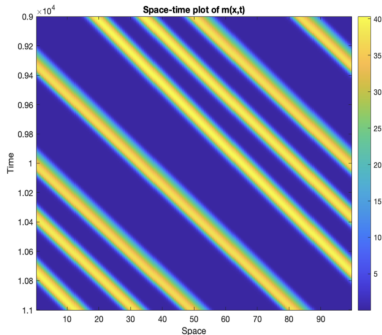


Figure 6: Representation of $m(x, t)$ for $(\chi, \tau) = (2, 100)$ and position in the bifurcation diagram.

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Why is this model useful?

Let's compare some $2D$ simulations with clinical data.

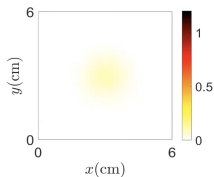
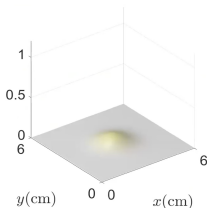
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$$\chi \ll \chi_c$$

$\Rightarrow$ homogeneous solution

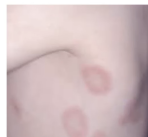
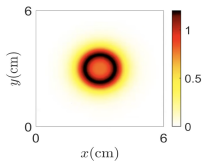
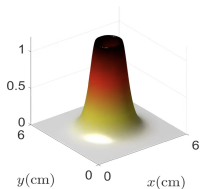
Macrophages
Day 1



$$L < 0, \quad \chi > \chi_c$$

$\Rightarrow$ ring solution

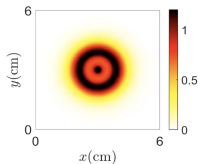
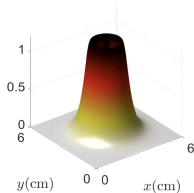
Macrophages
Day 40



$$L > 0, \quad \chi \gg \chi_c$$

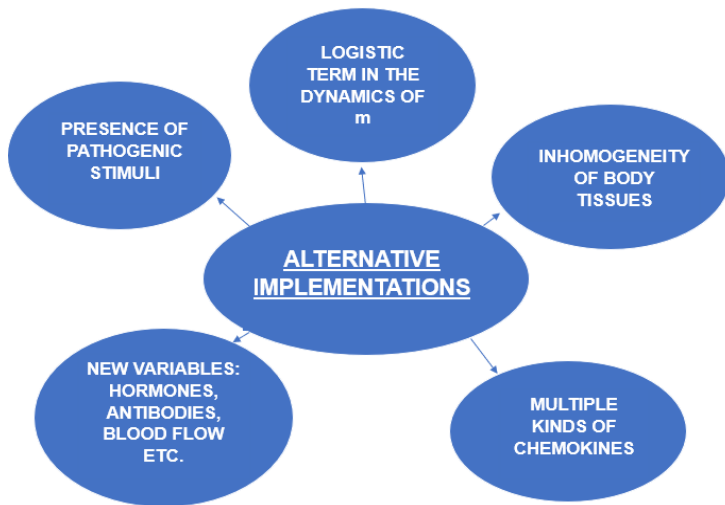
$\Rightarrow$ bump solution

Macrophages
Day 40



How could the model be improved?

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Thank you for your attention.

"Like the dynamic patterns in inflammation, may our knowledge continue to evolve and spread..."